

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	65.0	Gender	Male
Height (cm)	170.0	Weight (kg)	110.0
BMI	38.0	Product Info 2	D3
Occupation	Construction	Annual Income	\$45,000.00
Insurance Type	Life	Coverage Amount	\$200,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 8 of 8
Risk Category	High Risk
Recommended Annual Premium	\$40,000.00

3. AI Actuarial Analysis

Actuarial Evaluation Report: Insurance Application Review

Date: October 26, 2023

Applicant Details:

* **Age:** 65.0

* **Height:** 170.0 cm, **Weight:** 110.0 kg, **BMI:** 38.0

* **Product Info:** D3 (Specific product variant)

* **Occupation:** Construction

* **Income:** \$45,000.00

* **Smoker:** Yes (1)

* **Previous Claims:** 2

* **Family History:** Present (1)

* **Insurance Type:** Life

* **Coverage Amount:** \$200,000.00

* **Exercise:** No (0)

* **Alcohol:** Moderate (2)

* **Gender:** Male

Machine Learning Classifier Output:

* **Predicted Risk Class:** 8 (Highest Risk out of 8)

* **Prediction Probability/Confidence:** 33.0%

* **Calculated Annual Premium:** \$40,000.00

1. Risk Assessment

The applicant presents a highly elevated risk profile across multiple critical dimensions:

* **Age (65):** Significantly increases mortality risk due to advanced age.

* **BMI (38.0):** Classified as Obese Class II, presenting substantial health risks including cardiovascular disease, diabetes, and certain cancers, which are critical for life insurance.

* **Smoking Status (Smoker=1):** A primary risk factor, drastically increasing the likelihood of various diseases and premature mortality.

* **Occupation (Construction):** A physically demanding and often hazardous occupation, increasing the risk of accidental death or injury.

* **Previous Claims (2):** Indicates a history of insurance claims, suggesting higher future claim propensity or pre-existing health/risk issues that have manifested. This is a significant red flag.

* **Family History (1):** The presence of adverse family medical history further elevates genetic predisposition to certain conditions.

* **Lifestyle (Exercise=0, Alcohol=2):** Lack of exercise combined with moderate alcohol consumption, alongside smoking and obesity, points to a generally unhealthy lifestyle exacerbating health risks.

* **Coverage Amount vs. Income:** A coverage amount of \$200,000 on an annual income of \$45,000 represents a high income replacement ratio (approx. 4.4x), which, while not immediately excessive, requires careful consideration given the other high-risk factors.

In summary, the confluence of advanced age, severe obesity, smoking, hazardous occupation, prior claims history, and poor lifestyle places this applicant in a very high-risk category.

2. Comparative Analysis

The current application's risk profile (Risk Class 8, Premium \$40,000) is substantially higher than the provided similar historical approved cases:

* **POL-2626 (Risk Class 2, Premium \$12,000):** This case had a much lower risk profile. While the similarity (42.8%) suggests some shared characteristics (potentially insurance type or general demographics), the current applicant's severe health and lifestyle factors (BMI 38, smoker, 2 previous claims, age 65) fundamentally differentiate them, justifying a vastly higher risk classification and premium.

* **POL-25755 (Risk Class 5, Premium \$20,000):** This client was also approved at a moderate-to-high risk class. The current applicant's profile surpasses this level due to the aggregation of extreme risk factors (e.g., higher BMI, smoking, multiple claims), pushing them into a higher risk bracket.

* **POL-31852 (Risk Class 7, Premium \$30,000):** This case represents the highest risk among the comparables. While closer to the current application, the applicant's profile is even more adverse, particularly with the combination of a BMI of 38, active smoking, and two previous claims, which are likely more severe than the factors contributing to Risk Class 7 in the historical case. The current premium of \$40,000 reflects this incremental increase in severity beyond Risk Class 7.

The "similarity" metric from the RAG context, while helpful for identifying potentially related cases, does not fully capture the *magnitude* of the difference in risk factors present in this applicant. The current applicant's specific combination of high-impact risk factors is significantly more adverse than any of the approved historical cases, leading to the highest possible risk class.

3. Anomaly & Fraud Detection

Several factors warrant closer inspection:

* **High Previous Claims (2):** Two previous claims are significant and should be thoroughly investigated. The nature, timing, and severity of these claims are critical to understanding the underlying risk and ruling out potential non-disclosure or adverse selection.

* **Very High BMI (38.0) combined with Smoking and Lack of Exercise:** This combination points to severe health deterioration and poor lifestyle choices. While not an "anomaly" in terms of inconsistency, it's an extreme presentation of high-risk factors that significantly elevates mortality risk.

* **Affordability of Premium:** An annual premium of \$40,000 on an income of \$45,000 is practically unaffordable. This raises concerns regarding financial justification and potential moral hazard. It suggests the applicant may not intend to maintain the policy or that the coverage amount is not commensurate with their financial standing given the required premium. This requires detailed financial underwriting.

* **Low Prediction Confidence (33.0%):** The low confidence score from the ML classifier (33.0%) despite assigning the highest risk class (8) indicates that the model is not highly certain about this specific prediction. This could be due to the applicant being an outlier with an extreme combination of risk factors, or it could suggest data sparsity for such extreme profiles, leading to model uncertainty. This low confidence is a significant flag for manual review.

4. Premium Recommendation

The calculated annual premium of \$40,000.00 is aligned with the extreme risk profile identified.

* **Highest Risk Class:** The assignment to Risk Class 8 (the highest possible) directly correlates with the severe combination of advanced age, high BMI, smoking status, hazardous occupation, previous claims, and poor lifestyle. Each of these factors independently drives up the premium, and their combination leads to a multiplicative effect.

* **Historical Comparison:** The \$40,000 premium is higher than even the Risk Class 7 case (\$30,000), further affirming that the applicant's risk factors are considered more severe than any similar approved historical data points.

* **Model's Uncertainty:** While the premium is high, the low prediction confidence (33.0%) suggests the model might be pushing a conservative estimate due to the novelty or extremity of this risk profile. This reinforces the need for manual validation of the premium and the risk class.

Given the extreme risk factors, the premium in isolation appears actuarially justified in terms of mortality and morbidity expectation. However, its practical affordability and the low model confidence introduce complexities.

5. Underwriting Decision Recommendation

Recommendation: Refer for Manual Underwriting Review

Justification:

Despite the ML classifier assigning the highest risk class (8) and a corresponding high premium, several critical factors necessitate a thorough manual review by an experienced underwriter:

1. **Extreme Risk Profile:** The confluence of severe risk factors (age 65, BMI 38, smoker, construction, 2 previous claims, poor lifestyle) represents an exceptionally high mortality risk that requires granular assessment beyond automated classification.
2. **Low Prediction Confidence (33.0%):** The model's low confidence level indicates that this case falls outside its highly confident classification boundaries. A human underwriter can apply expert judgment, potentially identify non-obvious factors, and validate the model's output.
3. **Financial Justification & Affordability:** The high premium (\$40,000) relative to the applicant's income (\$45,000) raises significant concerns about affordability, persistency, and potential moral hazard. A detailed financial needs analysis and justification are imperative.
4. **Previous Claims:** The details of the two previous claims must be thoroughly investigated to understand their nature, impact on current health, and implications for future risk.
5. **Occupational Hazards:** A deeper dive into the specific duties within 'Construction' may be required, alongside any safety measures or associated risks.

A manual review will allow for a comprehensive assessment of all risk factors, clarification of the previous claims, thorough financial underwriting, and potentially the imposition of specific terms or exclusions, or a final decision to reject if the risk is deemed uninsurable at an acceptable premium.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$40,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.